Test_2

1. Conditions (if, elif, else)

20/20

1. Write a program to check whether a number is positive, negative, or zero.

```
num = int(input("Enter Number : "))

if num>0:
    print(f"{num} is positive")
elif num<0:
    print(f"{num} is Negative")
else:
    print(f"{num} is zero")

Enter Number :  3

3 is positive
```

1

2. Write a program to find the largest of three numbers.

```
num1 = int(input("Enter Number : "))
num2 = int(input("Enter Number : "))
num3 = int(input("Enter Number : "))

if num1>num2 and num1>num3:
    print(f"{num1} is Largest ")
elif num2>num3 and num2>num1:
    print(f"{num2} is Largest ")
elif num3>num1 and num3>num2:
    print(f"{num3} is Largest ")
else:
    print("All are Equal ")


Enter Number :  12
Enter Number :  12
Enter Number :  12

All are Equal
```

1

3. Write a program to check whether a given year is a leap year.

```
year = int(input("Enter the Year : "))

if year%400 == 0:
    print(f"{year} is leap year")
elif year%100 == 0:
    print(f"{year} is not leap year")
elif year % 4==0:
    print(f"{year} is leap year")
else:
    print(f"{year} is not leap year")

Enter the Year :  2000
```

1

```
2000 is leap year
```

4. Write a program to check if a character is a vowel or consonant.

```
ch = input("Enter a character : ")
vow="aeiouAEIOU"

if ch in vow:
    print(f"{ch} is vowel ")
else:
    print(f"{ch} is consonant")
```

```
Enter a character :  w

w is consonant
```

1

5. Write a program to calculate the electricity bill:

- First 100 units → ₹5/unit
- Next 100 units → ₹7/unit
- Above 200 units → ₹10/unit

```
unit = int(input("Enter a consumed unit : "))

if unit<=100:
    bill = unit * 5
    print("Bill Amount is ",bill)
elif unit>100 and unit<=200:
    bill = unit * 7
    print("Bill Amount is ",bill)
else:
    bill = unit * 10
    print("Bill Amount is ",bill)
```

```
Enter a consumed unit :  210

Bill Amount is  2100
```

1

2. Loops (for / while)

1. Print all even numbers from 1 to 100.

```
for i in range(1,101):
    if i%2==0:
        print(i,end = " ")
```

```
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48
50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 90 92 94 96
98 100
```

1

2. Find the factorial of a given number using a loop.

```
num = int(input("Enter a Number : "))

fact = 1
for i in range(1,num+1):
    fact = fact*i
```

1

```
print("Factorial is ",fact)
```

Enter a Number : 5

Factorial is 120

3. Print the Fibonacci series up to n terms.

```
num = int(input("Enter a Number : "))

a=0
b=1

for i in range(1,num+1):
    print(a,end =" ")
    a,b=b,a+b
```

Enter a Number : 7

0 1 1 2 3 5 8

4. Find the sum of digits of a given number.

```
num = int(input("Enter a Number : "))
temp = num
sum1 =0
while temp>0:
    rem = temp%10
    sum1 = sum1 + rem
    temp =temp//10
print(sum1)
```

Enter a Number : 1234

10

5. Check whether a number is an Armstrong number

```
num = int(input("Enter a Number :" ))

digit =len(str(num))
temp = num
sum1 = 0
while temp>0:
    digits = temp%10
    sum1 = sum1+ digits**digit
    temp =temp//10
if num == sum1:
    print("Armstrong")
else:
    print("Not Armstrong")
```

Enter a Number : 123

Not Armstrong

3. Functions

1

1

1

1. Write a function to find the greatest of three numbers.

```
def greatest(num1,num2,num3):
    if num1>num2 and num1>num3:
        print(f"{num1} is Largest ")
    elif num2>num3 and num2>num1:
        print(f"{num2} is Largest ")
    elif num3>num1 and num3>num2:
        print(f"{num3} is Largest ")
    else:
        print("All are Equal ")

greatest(30,40,20)

40 is Largest
```

1

2. Write a function to check whether a number is prime.

```
def prime(n):
    is_Prime=True

    if n<=1:
        print(f"{n} is not Prime")

    for i in range(2,n):
        if n%i == 0:
            is_Prime = False

    if is_Prime:
        print(f"{n} is Prime")
    else:
         print(f"{n} is not Prime")

n = int(input("Enter a Number :"))
prime(n)


Enter a Number : 6

6 is not Prime
```

1

1 x
2 → 1,2
3 → 1,3
4 x → 1,2,4 x
5 → 1,5
6
7 → 1,7

3. Write a function that returns the reverse of a string.

```
def rev(str1):
    st =""
    for i in str1:
        st = i+st
    print(st)

str1 = input("Enter a Number :")
rev(str1)

Enter a Number : wer

rew
```

1

4. Write a function to calculate the area and perimeter of a rectangle.

```
def area_peri(l,b):
    area = l*b
    peri=2*(l+b)

    print("Area is ",area)
    print("Perimeter is ",peri)

l= int(input("Enter Length : "))
b = int(input("Enter Breadth : "))

area_peri(l,b)
```

```
Enter Length :  3
Enter Breadth :  4

Area is  12
Perimeter is  14
```

1

5. Write a function that accepts a list and returns the second largest element.

```
def s_lar(lst):
    maxi = lst[0]
    sl=lst[0]

    for i in lst:
        if i>maxi:
            sl = maxi
            maxi = i
        elif i>sl and i!=maxi:
            sl = i
    print("Second Largest is ",sl)



lst = [1,3,4,5,7,8,34,98]
s_lar(lst)
```

```
Second Largest is  34
```

1

4. Exception Handling

1. Write a program to handle ZeroDivisionError.

```
try:
    a = int(input("Enter a Number 1"))
    b = int(input("Enter a Number 2"))

    print(a/b)
except ZeroDivisionError:
    print("Cannot Divide by Zero ")
except ValueError:
    print("Enter Valid Number")
```

```
Enter a Number 1 23
Enter a Number 2 0
```

1

```
Cannot Divide by Zero
```

2. Write a program to handle ValueError when the user enters invalid input.

```
try:
    a = int(input("Enter a Number 1: "))

    print(a)
except ValueError:
    print("Enter Valid Number")

Enter a Number 1:  s

Enter Valid Number
```

1

3. Create a custom exception to check if age is less than 18.

```
class AgeError (Exception):
    pass

try:
    age = int(input("Enter Your Age : "))
    if age<18:
        raise AgeError("Invalid Age, Please Enter age above 18 ")

except AgeError as e:
    print(e)
else:
    print("Eligible")
finally:
    print("Thank You")

Enter Your Age :  12

Invalid Age, Please Enter age above 18
Thank You
```

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4. Write a program that handles both IndexError and KeyError.

```
try:
    lst = [10, 20, 30]
    print(lst[5])

except IndexError:
    print("Index Out Of Range")

try:
    d = {"name": "Rahul"}
    print(d["age"])

except KeyError:
    print("Key Not Found")

Index Out Of Range
Key Not Found
```

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(indexError, KeyError)

5. Write a program using try, except, else, and finally to read a file and display its contents.

```
try:
    file = open("D:\\Python_Test_Questions-1.txt",'r')
except FileNotFoundError:
    print("File Name Error! Please Enter a valid file Name.")
else:
    print("Output is:\n",file.read())
finally:
    print("Thank You")
```

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```
Output is:
 Python Test Questions

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or
    zero.
2.  Write a program to find the largest of three numbers.
3.  Write a program to check whether a given year is a leap year.
4.  Write a program to check if a character is a vowel or consonant.
5.  Write a program to calculate the electricity bill:
    -   First 100 units â†’ â‚¹5/unit
    -   Next 100 units â†’ â‚¹7/unit
    -   Above 200 units â†’ â‚¹10/unit

2. Loops (for / while)

1.  Print all even numbers from 1 to 100.
2.  Find the factorial of a given number using a loop.
3.  Print the Fibonacci series up to n terms.
4.  Find the sum of digits of a given number.
5.  Check whether a number is an Armstrong number.

3. Functions

1.  Write a function to find the greatest of three numbers.
2.  Write a function to check whether a number is prime.
3.  Write a function that returns the reverse of a string.
4.  Write a function to calculate the area and perimeter of a
rectangle.
5.  Write a function that accepts a list and returns the second
largest
    element.

4. Exception Handling

1.  Write a program to handle ZeroDivisionError.
2.  Write a program to handle ValueError when the user enters invalid
    input.
3.  Create a custom exception to check if age is less than 18.
4.  Write a program that handles both IndexError and KeyError.
5.  Write a program using try, except, else, and finally to read a
file
    and display its contents.
```

```
Code Executed Successfully
```